

Premise: I analyzed lyrics written by Olivia Rodrigo and Gracie Abrams based on how well they fit the seven basic plot archetypes, and recorded the data. My goal is to determine which archetypes are the best classifiers of each artist, and for each individual album I reviewed in the data.

Implementation Plan: I set up a logistic regression model to classify the songs as either written by Olivia Rodrigo (class 0) or Gracie Abrams (class 1). The features recorded for each song were the rating of fit of the lyrics to each of the seven plots, on a scale of 0-3. These features were scaled to follow a standard normal distribution so the magnitudes of the coefficients could be compared. Since this is a binary classifier, features with large positive coefficients are good indicators of Gracie Abrams, and features with large negative coefficients are good indicators of Olivia Rodrigo.

However, I also want to set up a logistic regression model to classify the songs according to the albums they belong to. A multinomial softmax model is ideal for this task because the dataset is small (around 90 total songs) and the classes are mutually exclusive (each song belongs to only one album). Olivia Rodrigo's SOUR album will be treated as the base class, and each other album will be compared against it. Positive coefficients mean the song is less likely to belong to the base class, and vice versa. This doesn't directly give me a profile for each album, but I can analyze how each album differs from SOUR to compare them to each other.